Deep Margin Cosine Autoencoder-Based Medical Hyperspectral Image Classification for Tumor Diagnosis

Meiling Wang, Yongchang Xu, Zhisheng Wang, and Changda Xing

***Abstract*— Deep learning has become a powerful tool to automatically classify medical hyperspectral images (MedHSIs) for the diagnosis of various tumors, such as cancer. These deep-learning-based classification approaches consist of both feature extraction and disease prediction, which are independent of each other. Therefore, the extracted features may be incompatible with the used classifier for prediction. To remedy such deficiency, in this work, we propose a novel deep method, i.e., deep margin cosine autoencoder (DMCA), for the MedHSI classification, which provides solid support to diagnose tumors. To be specific, *first*, we consider deep autoencoder (DAE) network as the basic framework for the feature extraction of the MedHSI, and the soft-max classifier is introduced as the output layer to predict the results. *Second*, to report the features well compatible with the classifier, the importance of the soft-max is added as a constraint into the DAE network. Furthermore, a cosine margin is introduced to enhance the discrimination of different feature clusters. In addition, we also design an optimization scheme to the solutions of the DMCA model. *Third*, a two-stage training strategy is presented to train the built DMCA network. After completing the training, unknown tissues (i.e., unlabeled samples) can be specified to determine whether they are tumors or not. Sufficient experimental results have been provided to validate that our DMCA method achieves better classification performance and higher tumor diagnosis accuracy compared with some advanced approaches.**

***Index Terms*— Deep learning, deep margin cosine autoencoder (DMCA), medical hyperspectral image (MedHSI) classification, tumor diagnosis, two-stage training strategy.**

I. Introduction

TUMOR diseases, such as cancer, have extremely high incidence and death rates, which are a serious public health problem and threaten the life of numerous people. Accurate tumor diagnosis has played an important role in timely treatment and increasing the probability of successful cure and has attracted high attention from the global scientific community. Presently, the gold standard of tumor diagnosis is still histopathology. For instance, Li et al. [1] presented a generalized framework of feature learning-enhanced convolutional neural network (CNN) for pathology-image-oriented cancer diagnosis. However, this kind of diagnostic methods inevitably causes harm to the human body, and due to the subjective judgment of pathological experts, the diagnostic results may be one-sided. In addition, early screening and diagnosis of tumor commonly use detection methods, such as magnetic resonance imaging (MRI), X-ray images, and positron emission tomography–computed tomography (PET–CT). For example, Wu et al. [2] proposed an MRI-oriented novel attention-based glioma grading network (AGGN) with multiscale feature extraction and multimodal information fusion for tumor diagnosis. Li et al. [3] designed a computer-aided diagnosis model for accurate recognition of abnormal lesion from chest X-ray images. However, these imaging methods take a long time to detect and cannot locate in real time. Currently, there is an urgent clinical need to develop a safe, noncontact, and label-free imaging method to provide reliable support for real-time tumor resection during neurosurgery. Hyperspectral imaging is a fast, noncontact, nonionizing, noninvasive, high-safety, and label-free imaging method, which has shown great application potential in the field of biomedicine [4], [5], [6] and provides a new perspective to diagnose diseases, especially for malignant tumors, such as cancer. As a kind of advanced technique, the hyperspectral imaging has the strong ability to perceive the spatial resolution and spectral resolution, due to effectively capturing the radiance of materials from hundreds of contiguous narrow spectral bands. Therefore, it can well reflect the chemical composition and physical form information of body parts [7], [8].

At present, there are many typical medical hyperspectral image (MedHSI)-based diagnosis examples. Kumar et al. [9] presented an imaging system with principal component analysis (PCA) and Fourier transform infrared (FTIR) spectroscopy in order to diagnose breast cancer. The FTIR was used in histopathological specimens of breast cancer with various histological grades, which also reports spectral changes near carcinoma. The PCA was further utilized to analyze the collected data. Prominent features were observed in the 5882–6250-nm band, which could be exploited for cancer

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identification. Liu et al. [10] noninvasively measured and analyzed the reflectance spectra of tongue for tumor detection. Dicker et al. [11] introduced the routine hematoxylin and eosin (H&E) stained samples to distinguish malignant and benign dermal tissue in the spectral domain. In their findings, the differences of spectral signatures could be found when controlling the section thickness and staining time. For an abnormal body part, the collected MedHSI data usually digitally describe tumor, normal tissue, and other targets or background. To achieve intelligent disease diagnosis, the classification of MedHSIs is available to effectively identify the pathological tissues or structures. In addition, MedHSIs are actually a biomedical measurement result, which can simultaneously provide spatial and spectral information of biological tissues. They show diagnostic information about tissue physiology, morphology, and biochemical components. The classification task is the analysis of the information obtained from a set of hyperspectral measurements, which provide the supports for intelligent disease diagnosis. Therefore, this work contributes to the society of instrumentation and measurement in biomedical measurement.

The early studies of the MedHSI classification have mainly used original pixel vectors as the discriminative features, which are further processed via some popular classifiers, such as support vector machine (SVM) and logistic regression (LR) [12], [13], [14]. However, these works may suffer from the low classification accuracy due to the use of redundant spectral features. For the sake of obtaining more valuable spectral information, there currently exist many studies based on spectrum observation/analysis. For example, Verebes et al. [15] have used an enhanced dark-field microscope for the analysis of the spectral characteristics of blood cell compositions, so as to build spectral libraries to divide active cells from inactive cells. Conti et al. [16] have exploited eight scattering spectral libraries for the MedHSI classification. The work in [17] realizes the blind decomposition of the MedHSI via sparse learning. Pardo et al. [18] present that the directional kernel density estimation is applied into the classification of breast tissue spectra. Ravi et al. [19] have introduced manifold embedding and semantic segmentation into the analysis of MedHSIs used for brain observation. Wang et al. [20] have proposed a joint spatial–spectral analysis algorithm, which uses morphologic theories to complete the classification task of white blood cells.

In recent several years, deep learning [21] has provided a new manner to deal with the problem of the MedHSI classification. As a powerful tool, the deep learning has the strong ability to describe the intrinsic information hidden behind the MedHSI data and obtain an optimal decision boundary for the classification. Based on such competitive ability, the deep-learning-based methods [22], [23], [24], [25], [26], [27], [28], [29], [30] have the greater classification performance, compared with those traditional approaches. Halicek et al. [23] developed a CNN classifier to classify excised, squamous-cell carcinoma, thyroid cancer, and normal head and neck tissue samples included in MedHSI. The empirical results of 50 patients indicated that the MedHSI and CNN have the great potential for automatic tissue labeling of surgical specimens of head and neck patients. In [24], a deep CNN was employed to well realize the blood cell discrimination in MedHSI. In order to fully utilize spectral properties of 3-D MedHSI, Wang et al. [25] proposed a 3-D CNN for the classification, which is capable of effectively segmenting melanoma from the pathological MedHSI. More advanced, to remedy the deficiency of supervised CNN models suffering from the lack of samples, Wei et al. [26] proposed a deep autoencoder (DAE) with two channels for classifying blood cells, which learns joint spatial–spectral information in an unsupervised manner. Hao et al. [29] presented a method based on the fusion of multiple deep models (FMDM) for human brain MedHSI classification to identify glioblastoma tumor. Zeng et al. [30] provided a fusion transformer (FUST) with parallel CNN for microscopic MedHSI classification.

It is widely known that the hyperspectral image classification was originally used in the field of remote sensing, which aims to identify and distinguish the types of different ground targets [31], [32], [33]. Due to similar imaging mechanisms, a lot of classification approaches in remote sensing can be directly used for MedHSI [26], [34]. In the light of this, besides CNN and DAE, other deep-learning frameworks, such as generative adversarial network (GAN) [35], [36], [37] and deep belief network (DBN) [38], [39], [40], in the field of remote sensing can also be used for the MedHSI classification. These deep-learning-based methods have been empirically validated for their competitive performance, but they are confronted with a deficiency that may decrease the classification performance of MedHSI. More specifically, the deep-learning-based methods for the MedHSI classification generally consist of two parts, including both feature extraction and label prediction. The prediction part is achieved by a classifier-like soft-max at the top layer of the used deep network. These two parts are independent of each other. In other words, there is no correlation between the classifier and the feature extraction, so that the exported deep features may be incompatible with the classifier, resulting in unsatisfactory classification results.

In this article, we are devoted to developing a novel deep-learning method [i.e., deep margin cosine autoencoder (DMCA)] for the MedHSI classification, where the importance of the used classifier is considered into the process of feature extraction. The overall framework of the proposed DMCA method for MedHSI classification is shown in Fig. 1. To be specific, we first consider to introduce DAE network as the basic framework for the feature extraction of the MedHSI, and the soft-max classifier is used as the output layer to predict labels. Then, in order to yield the features well compatible with the classifier, the importance of the soft-max is added as a constraint into the DAE network-based feature learning model. Moreover, a cosine margin is introduced to enlarge the discrimination and separability of different feature classes. By such way, the DMCA network model is constructed. We also design an optimization algorithm to obtain the solutions of the DMCA model. Third, a two-stage training strategy is presented to train the built DMCA network. After completing the training, unknown tissues (i.e., unlabeled samples) can be specified to determine whether they are tumors or not. We evaluate the classification performance of the DMCA

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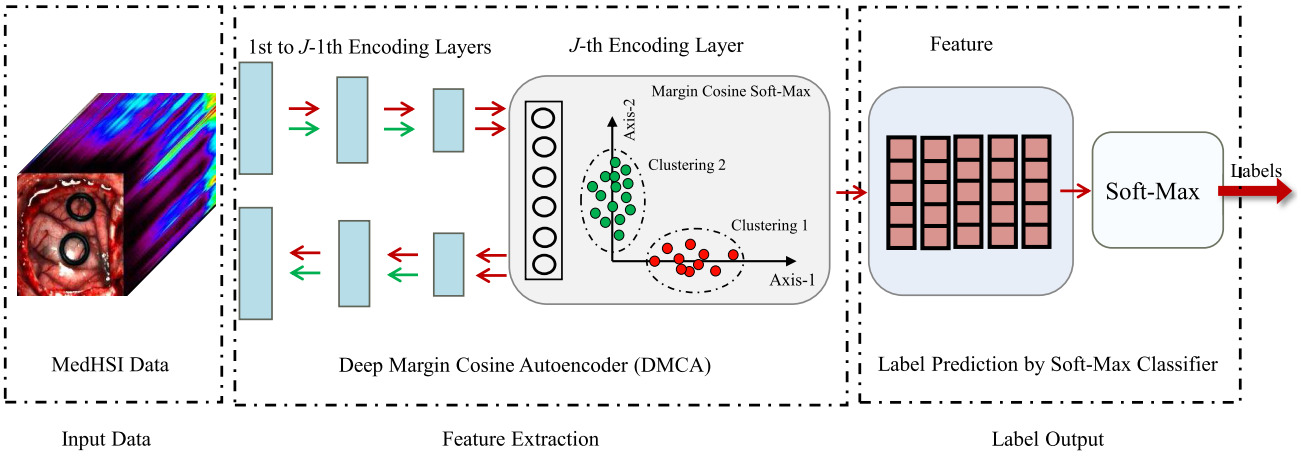


Fig. 1. Overall framework of the proposed DMCA method for MedHSI classification.

on two medical hyperspectral datasets, with empirical results demonstrating the effectiveness and superiority in the MedHSI classification task.

The major novel contributions of this article are summarized as the following twofold points. *First*, we propose to link the feature extraction and the classifier together and design a DMCA feature extraction method with the importance of classifier for MedHSI classification. To be specific, we embed the soft-max classifier as the constraint into the DAE network model for feature extraction of MedHSIs. By such way, a novel improved DAE network (i.e., DMCA) with the importance of the soft-max classifier is built for feature extraction to classify MedHSIs. *Second*, we introduce an extra cosine margin into the embedded soft-max, which can maximize the decision margin in the angular space, so that the extracted features tend to be more compact and separable.

The remainder of this article is organized as follows. The proposed DMCA method for MedHSI classification is introduced in Section II, including study motivation, model formulation, optimization algorithm, and training strategy. In Section III, experimental results on two medical hyperspectral datasets and related discussion are provided. We finally conclude this article in Section IV.

II. Methodology

We provide a novel DMCA method for the classification of MedHSIs in this work. Some notes are here shown as follows: $\mathbf{H} \in \mathbb{R}^{T\times O\times Q}$ denotes an input 3-D MedHSI, where T and $N = O \times Q$ are the spectral resolution (i.e., number of bands) and spatial resolution (i.e., size of each band), respectively. For convenient description, the given $\mathbf{H}$ is unfolded into its 2-D form $H \in \mathbb{R}^{T\times N}$.

A. Study Motivation and Model Formulation

Deep learning has become a powerful tool to complete the classification task of MedHSIs for disease diagnosis. Among general deep-learning forms, the DAE has the competitive ability of generalization and representation for the MedHSI data. The DAE form is composited of a plurality of autoencoders. In order to achieve label prediction, a classifier is added at the top layer of the DAE, which mostly uses the soft-max classifier for the discriminant calculation. In summary, the DAE-based classification methods are thought to consist of two parts, i.e., both feature extraction and label prediction. As can be known, these two stages are independent of each other. In other words, the importance of the used classifier does not participate in the stage of the feature extraction, so that the features exported by the DAE may be incompatible with the classifier, resulting in unsatisfactory label results.

According to the above description, extracting features independently of the classifier is not always optimal. The relative importance of features may sometimes depend on the type of classifier used [41]. This implies that good compatibility between feature extraction and classifiers is very valuable and potential to obtain great classification results. Based on such argument, incorporating the importance of the used classifier into the feature extraction process establishes the compatibility between the features and the classifier, which can improve classification performance. Therefore, for the sake of embracing the features that are more compatible with the classifier, we consider to introduce the importance of the classifier into the feature extraction, which is the *motivation* of this work. To be more specific, we attempt to construct a novel improved DAE network with the property of soft-max to extract the features and use the soft-max-based classifier to yield the labels of MedHSIs. We here recall the soft-max-based classifier, which has the corresponding loss form loss as follows:

$$\text{loss} = -\sum_{k=1}^{|\Omega|} \log p_k = -\sum_{k=1}^{|\Omega|} \log \frac{e^{W_k^T \mathbf{x}}}{\sum_{q=1}^{|\Omega|} e^{W_q^T \mathbf{x}}} \tag{1}$$

where Ω denotes the answer set; $\mathbf{x} \in \mathbb{R}^{L\times 1}$ is a given input sample vector; p_k denotes the posterior probability of $\mathbf{x}$ being correctly represented; $W \in \mathbb{R}^{L\times U}$ is the weight matrix. Let $W_k^T\mathbf{x} = \|W_k\|_2 \cdot \|\mathbf{x}\|_2 \cdot \cos(W_k, \mathbf{x})$, where $\cos(W_k, \mathbf{x})$ is the cosine of the angle between W_k and $\mathbf{x}$. The $L2$ normalization is further imposed on the weight W_k and feature vector $\mathbf{x}$ into a scale factor s, that is, $s = \|W_k\|_2 \cdot \|\mathbf{x}\|_2$. In the light of this,

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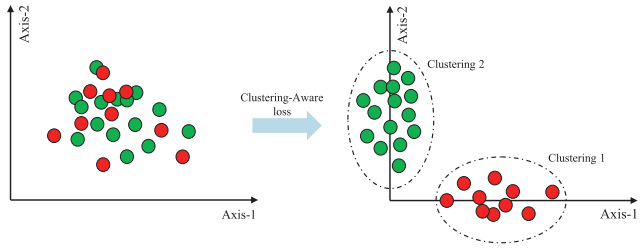


Fig. 2. Diagrammatic sketch to visually present the importance of the clustering-aware loss.

(1) is rewritten as follows:

$$\text{loss}_s = -\sum_{k=1}^{|\Omega|} \log \frac{e^{s\cos(W_k,\mathbf{x})}}{\sum_{q=1}^{|\Omega|} e^{s\cos(W_q,\mathbf{x})}}. \tag{2}$$

Equation (2) implies that both norm and angle of vectors can be contributed to the posterior probability. As can be further found, the loss in (2) is actually clustering-aware. More specifically, $\cos(W_k, \mathbf{x})$ reflects the angle between both W_k and $\mathbf{x}$, which promotes the exported features from the same class to be around an axis. Considering the case of $U = 2$ for example, sample $\mathbf{x}$ is mapped into the 2-D space, which is located near one of axes. If the feature vector of $\mathbf{x}$ is located near axis 1, there exists $\cos(W_1, \mathbf{x}) > \cos(W_2, \mathbf{x})$. On the contrary, $\cos(W_1, \mathbf{x}) < \cos(W_2, \mathbf{x})$. Here, in order to further improve the importance of such clustering-aware operation, a cosine margin m is introduced so as to satisfy $\cos(W_1, \mathbf{x}) - m > \cos(W_2, \mathbf{x})$ or $\cos(W_1, \mathbf{x}) < \cos(W_2, \mathbf{x}) - m$, where m is a positive constant. Because $\cos(W_k, \mathbf{x}) - m$ ($k = 1$ and 2) is smaller than $\cos(W_k, \mathbf{x})$, the constraint of the clustering is more stringent. For convenient understanding, a diagrammatic sketch is shown in Fig. 2 to visually present the importance of the above clustering-aware loss. As can be found from this figure, such loss constraint promotes the discrimination of features along axes. These analysis can be well generalized to the case of $U > 2$. Thus, the altered loss reinforces the clustering discrimination of learned features by encouraging an extra margin in the cosine space, which can be modeled as follows:

$$\text{loss}_{sm} = -\sum_{k=1}^{|\Omega|} \log \frac{e^{s(\cos(W_k,\mathbf{x})-m)}}{\sum_{q\neq k} e^{s\cos(W_q,\mathbf{x})} + e^{s(\cos(W_k,\mathbf{x})-m)}}. \tag{3}$$

Suggested by the above description, in this study, we build a novel DAE network with the soft-max classifier for feature extraction to classify MedHSIs, whose general model can be expressed as follows:

$$\arg\min \mathcal{F}(H, \varphi), \quad \text{s.t. } \text{loss}_{sm} \to 0 \tag{4}$$

where $\mathcal{F}(H, \varphi)$ is the DAE network and φ denotes the parameter set. Removing the constraint, (4) is equivalent to

$$\arg\min \mathcal{F}(H, \varphi) + \lambda_2 \text{loss}_{sm} \tag{5}$$

where λ_2 denotes the penalty parameter. As can be found, the DAE form $\mathcal{F}(H, \varphi)$ and the soft-max form loss_{sm} are linked by tuning the parameter λ_2. Here, the stacked autoencoder (SAE) is considered as the specific form of DAE, and the sparsity is imposed on the top layer; (5) can be rewritten as follows:

$$\begin{aligned}\arg\min_{H_j,P_j,W_j} &\sum_{j=1}^{J} \left\| H_j - P_j W_j^T H_j \right\|_F^2 + \lambda_1 \left\| W_J^T H_J \right\|_1 \\ &- \lambda_2 \sum_{n=1}^{N}\sum_{u=1}^{U} \log \frac{e^{s(\cos(W_{J_u},H_{J_n})-m)}}{\sum_{i\neq u} e^{s\cos(W_{J_i},H_{J_n})} + e^{s(\cos(W_{J_u},H_{J_n})-m)}}\end{aligned} \tag{6}$$

where λ_1 denotes the penalty parameter; J is the amount of deep layers; H_j, P_j, and W_j are the input, decoding matrix, and encoding matrix of the jth layer ($1 \le j \le J$), where $H_j = W_{j-1}^T H_{j-1}$ and $H_1 = H$; W_{J_l} is the lth column of the encoding matrix $W_J \in \mathbb{R}^{L\times U}$ at the Jth layer; H_{J_n} is the nth pixel vector of the input $H_J \in \mathbb{R}^{L\times N}$ at the Jth layer; $\cos(W_{J_l}, H_{J_n}) = W_{J_l}^T/(\|W_{J_l}\|_2) \cdot H_{J_n}/(\|H_{J_n}\|_2)$.

We remark that (6) is called DMCA, which integrates the DAE and the margin cosine soft-max loss to extract features for MedHSI classification. The first component in (6) is the DAE with J layers, which is used to mine and extract the deep features of the given MedHSI H. The second one is the sparse regularizer so as to guarantee the sparsity of the extracted features. The third component is the margin cosine classifier term, which not only reflects the importance of the soft-max classifier, but also maximizes the decision margin in the angular space, so that the extracted features can be easily and conveniently categorized by the soft-max.

As can be observed in formulation (6), the $L1$-based regularizer and the margin cosine term are actually only related to the deepest layer of the first component, that is,

$$\begin{aligned}\arg\min_{H_j,P_j,W_j} &\sum_{j=1}^{J-1} \left\| H_j - P_j W_j^T H_j \right\|_F^2 \\ &+ \left\| H_J - P_J W_J^T H_J \right\|_F^2 + \lambda_1 \left\| W_J^T H_J \right\|_1 \\ &- \lambda_2 \sum_{n=1}^{N}\sum_{u=1}^{U} \log \frac{e^{s(\cos(W_{J_u},H_{J_n})-m)}}{\sum_{i\neq u} e^{s\cos(W_{J_i},H_{J_n})} + e^{s(\cos(W_{J_u},H_{J_n})-m)}}.\end{aligned} \tag{7}$$

In (7), the solutions of variables H_j, P_j, and W_j ($1 \le j \le J-1$) can be obtained by training the DAE. The H_J can be computed by $H_J = W_{J-1}^T H_{J-1}$, while P_J and W_J can be optimized via the following form:

$$\begin{aligned}\arg\min_{P_J,W_J} &\left\| H_J - P_J W_J^T H_J \right\|_F^2 + \lambda_1 \left\| W_J^T H_J \right\|_1 \\ &- \lambda_2 \sum_{n=1}^{N}\sum_{u=1}^{U} \log \frac{e^{s(\cos(W_{J_u},H_{J_n})-m)}}{\sum_{i\neq u} e^{s\cos(W_{J_i},H_{J_n})} + e^{s(\cos(W_{J_u},H_{J_n})-m)}}.\end{aligned} \tag{8}$$

This form can be seen as the uniformed layer, which integrates the deepest sparse layer of the autoencoder and the margin cosine term together. Next, we will show an optimization algorithm to solve (8).

B. Optimization Algorithm

In (8), there exist variables P_J and W_J. It should be noted that the $L1$-norm computation for H_J is included. In order

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to obtain the optimal H_J, an auxiliary variable $\mathcal{A} = W_J^T H_J$ is here introduced to relax (8). Thus, (8) can be further transformed into

$$\begin{aligned}&\arg\min_{P_J, W_J, \mathcal{A}} \left\| H_J - P_J W_J^T H_J \right\|_F^2 + \lambda_1 \|\mathcal{A}\|_1 \\ &\quad - \lambda_2 \sum_{n=1}^{N} \sum_{u=1}^{U} \log \frac{e^{s\left(\cos\left(W_{J_u}, H_{J_n}\right) - m\right)}}{\sum_{i \neq u} e^{s \cos\left(W_{J_i}, H_{J_n}\right)} + e^{s\left(\cos\left(W_{J_u}, H_{J_n}\right) - m\right)}} \\ &\text{s.t. } \mathcal{A} = W_J^T H_J. \end{aligned} \tag{9}$$

Here, define a diagonal matrix $M = \text{diag}(m, m, \ldots, m) \in \mathbb{R}^{U \times U}$. Removing the constrained condition, we further rewrite (9) into

$$\begin{aligned}&\arg\min_{P_J, W_J, \mathcal{A}} \left\| H_J - P_J W_J^T H_J \right\|_F^2 + \lambda_1 \|\mathcal{A}\|_1 + \eta \left\| \mathcal{A} - W_J^T H_J \right\|_F^2 \\ &\quad - \lambda_2 \sum_{n=1}^{N} \sum_{u=1}^{U} \log \frac{e^{s\left(\cos\left(W_{J_u}, H_{J_n}\right) - M_{u,u}\right)}}{\sum_{i=1}^{U} e^{s\left(\cos\left(W_{J_i}, H_{J_n}\right) - M_{i,u}\right)}} \end{aligned} \tag{10}$$

where η is a coupling parameter. An optimization algorithm with alternating iterative updating is here designed to iteratively optimize (10) to minimize the loss of one variable when fixing other variables. More specifically, the following hold.

1) Updating P_J: When the variables W_J and $\mathcal{A}$ are fixed, the P_J can be updated by

$$\arg\min_{P_J} \left\| H_J - P_J W_J^T H_J \right\|_F^2. \tag{11}$$

Equation (11) has a closed-form solution as follows:

$$P_J = H_J H_J^T W_J \left(W_J^T H_J H_J^T W_J \right)^{-1}. \tag{12}$$

2) Updating W_J: Once the variables P_J and $\mathcal{A}$ are fixed, the W_J is updated via

$$\begin{aligned}&\arg\min_{W_J} \left\| H_J - P_J W_J^T H_J \right\|_F^2 + \eta \left\| \mathcal{A} - W_J^T H_J \right\|_F^2 \\ &\quad - \lambda_2 \sum_{n=1}^{N} \sum_{u=1}^{U} \log \frac{e^{s\left(\cos\left(W_{J_u}, H_{J_n}\right) - M_{u,u}\right)}}{\sum_{i=1}^{U} e^{s\left(\cos\left(W_{J_i}, H_{J_n}\right) - M_{i,u}\right)}}. \end{aligned} \tag{13}$$

Equation (13) can be rewritten into

$$\begin{aligned}&\arg\min_{W_{J_u}} \sum_{u=1}^{U} \left(\left\| H_J - P_{J_u} W_{J_u}^T H_J \right\|_F^2 + \eta \left\| \mathcal{A} - W_{J_u}^T H_J \right\|_F^2 \right) \\ &\quad - \lambda_2 \sum_{n=1}^{N} \sum_{u=1}^{U} \log \frac{e^{s\left(\cos\left(W_{J_u}, H_{J_n}\right) - M_{u,u}\right)}}{\sum_{i=1}^{U} e^{s\left(\cos\left(W_{J_i}, H_{J_n}\right) - M_{i,u}\right)}}. \end{aligned} \tag{14}$$

We here seek the derivative of the objective function [denoted by $\mathcal{L}(W_J)$] in (14) with regard to W_{J_u} as the following:

$$\begin{aligned}&\frac{\partial \mathcal{L}(W_J)}{\partial W_{J_u}} \\ &= 2\left(H_J H_J^T W_{J_u} P_{J_l}^T P_{J_l} - H_J H_J^T P_{J_l} \right) \\ &\quad + 2\left(H_J H_J^T W_{J_u} - H_J \mathcal{A}^T \right) \\ &\quad + \lambda_2 \sum_{n=1}^{N} \left(\left(\frac{L e^{s\left(\cos\left(W_{J_u}, H_{J_n}\right) - M_{u,u}\right)}}{\sum_{i=1}^{U} e^{s\left(\cos\left(W_{J_i}, H_{J_n}\right) - M_{i,u}\right)}} - 1 \right) \times s \right. \\ &\qquad \left. \times \frac{\|W_{J_u}\|_2 I - W_{J_u} W_{J_u}^T}{\|W_{J_u}\|_2^3} \times \frac{H_{J_n}}{\|H_{J_n}\|_2} \right). \end{aligned} \tag{15}$$

With the gradient descent strategy, we acquire the updated W_{J_u} by

$$W_{J_u} = W_{J_u} - \kappa \frac{\partial \mathcal{L}(W_J)}{\partial W_{J_u}} \tag{16}$$

where κ is the learning rate. Once all $W_{J_u} (1 \leq u \leq U)$ is obtained, the update W_J will be expressed into $W_J = [W_{J_1}, W_{J_2}, \ldots, W_{J_U}]$.

3) Updating $\mathcal{A}$: Keeping variables P_J and W_J fixed, the subproblem with respect to $\mathcal{A}$ can be acquired as follows:

$$\arg\min_{\mathcal{A}} \lambda_1 \|\mathcal{A}\|_1 + \eta \left\| \mathcal{A} - W_J^T H_J \right\|_F^2. \tag{17}$$

Equation (17) yields the closed-form solution by using the soft-thresholding scheme, that is,

$$\mathcal{A} = \text{soft}\left(W_J^T H_J, \frac{\lambda_1}{2\eta} \right) \tag{18}$$

where $\text{soft}(a, b) = \text{sign}(a) \times \max(|a| - b, 0)$ and $\text{sign}(a)$ denotes the sign function.

Algorithm 1 Optimization Algorithm to Calculate the Solutions W_J and P_J of the Jth Layer (8)

Input: Current layer input H_J; Parameters λ_1, λ_2, and η; Learning rate κ.
Output: Encoding matrix W_J; Decoding matrix P_J.
Optimization Procedure:
Initialize $W_J \in \mathbb{R}^{L \times U}$; $\mathcal{A} \in \mathbb{R}^{U \times N}$
while not converge do
  Update P_J by (12);
  for u=1:U
    while not converge do
      Obtain W_{J_u} by (15) and (16);
    end while converge
  end for
  Update $W_J = [W_{J_1}, W_{J_2}, \ldots, W_{J_U}]$;
  Update $\mathcal{A}$ by (18);
end while converge
return W_J and P_J.

When the convergence condition $\|\mathcal{A} - W_J^T H_J\|_F^2 \to 0$ (i.e., $\|\mathcal{A} - W_J^T H_J\|_F^2 \leq \kappa$, where κ is a small positive threshold value and set to 10^{-3}), the above threefold iterative updating operations will be stopped, and the final optimal W_J will be obtained. The algorithm flow of the designed optimization algorithm to compute the solutions W_J and P_J of the Jth layer (8) is shown in Algorithm 1. Here, an example is exploited to visually show the evolution of the cost function of our DMCA to observe our optimization strategy in Fig. 3. As can be observed, after undergoing a small amount of fluctuation during the iteration range of $[0, 5]$, the value of the cost function of our DMCA is monotonically decreasing with the increasing of iteration times, until the convergence is almost reached at about 20 iterations. That is, the cost function is minimized after 20 iterations. It can be verified that our proposed optimization algorithm is effective and efficient.

C. Two-Stage Training Strategy

In this section, we give a training strategy for the proposed DMCA network. Here, assume $H \in \mathbb{R}^{T \times N}$ as the set of

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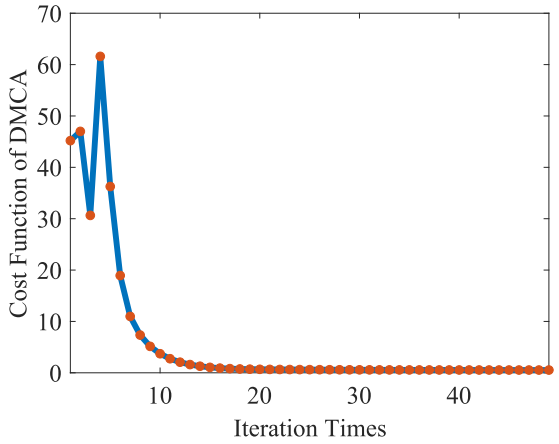


Fig. 3. Evolution of the cost function of DMCA on iteration times.

training samples with label $L_H \in \mathbb{R}^{1\times N}$, where T is the spectral resolution and N is the number of training samples. Define $H \in \mathbb{R}^{T\times N'}$ as unlabeled MedHSI data with N' sample vectors. The trained network is set to $J+1$ layers, where the first J layers are used to encode the input data and the final single layer (i.e., $(J+1)$th layer) is of soft-max for label output.

On the side of training, we present a two-stage strategy to train the proposed DMCA network. Specifically, we first exploit training sample set H with label L_H to train the DMCA form without the Jth layer, where the first $J-1$ layers and the $(J+1)$th layer are here included as a kind of traditional DAE-based prediction network. By this way, the parameters of the first $J-1$ encoding layers and $(J+1)$th layer can be initialized, which is the *first training stage* of our two-stage strategy. In the *second training stage*, we independently train the Jth layer [i.e., the model in (8)] to determine the optimal W_J and P_J. The training data H_J are the output of the $(J-1)$th layer, which can be obtained by the first training stage.

On the side of prediction, once the proposed DMCA network has been trained, the extracted feature form F of MedHSIs can be described as $F = W_J^T E_J$, where E_J is the export of the trained first $(J-1)$th layers for unlabeled MedHSI E. The final labels are predicted by adding the extracted feature F into the soft-max layer.

III. Experiment

A. Experimental Data Preparation

In order to evaluate the performance of the proposed DMCA classification approach, two medical hyperspectral datasets are exploited, including *In Vivo Hyperspectral Human Brain (In Vivo HHB)* database [42], [43] and *Multidimensional Choledoch (MC)* database [44], [45]. The In Vivo HHB dataset was generated by using a customized intraoperative hyperspectral acquisition system, delineating human brain tumors in real time during neurosurgical operations. The imaging camera is based on a push-broom technique and covers the spectral range from 400 to 1000 nm with a spectral resolution of 2–3 nm, which can capture 826 spectral bands. This dataset consists of 36 MedHSIs from different patients, and each MedHSI

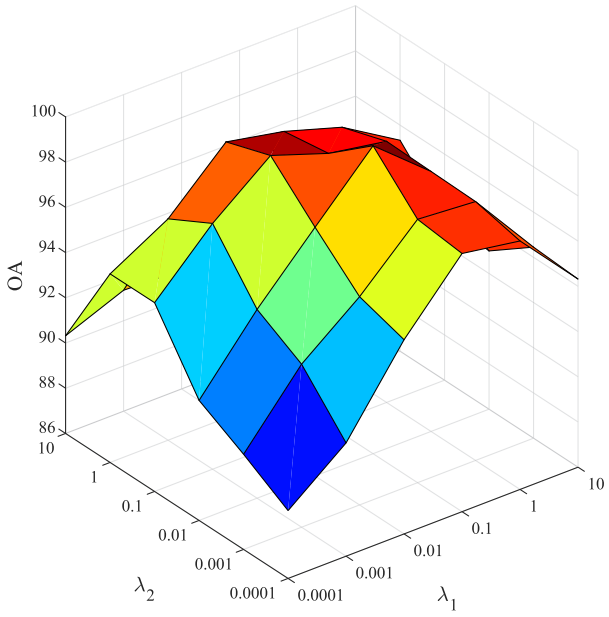


Fig. 4. Performance (in OA) of DMCA affected by parameters λ_1 and λ_2.

was defined with four labeled classes, including normal tissue, tumor tissue, blood vessel, and background elements. The MC dataset was obtained via a microscopy hyperspectral imaging system, which includes MedHSIs of choledoch tissues and was provided by Changhai Hospital, Shanghai, China. This dataset contains 880 scenes of multidimensional MedHSIs captured from choledoch tissues of 174 patients. Among these MedHSIs, 689 scenes reflect part of cancer areas, 49 scenes are full of cancer areas, and 142 scenes are images without cancer areas. For each MedHSI, it contains 60 bands and has the spatial size of 1280 $\times$ 1024.

B. Quantitative Metrics

In our experimental studies, in order to quantitatively evaluate the quality of classification results, three metrics, including overall accuracy (OA), average accuracy (AA), and Kappa coefficient, are utilized. Among these metrics, the OA index denotes the percentage of pixels classified properly. The AA represents the average percentage of pixels included in each class. The Kappa corrects the correct percentage of classified pixels. In addition, we count the correct identification rate for each class, which can individually reflect classified pixels from single class.

C. Analysis of Parameter Section

In experiments, we use the grid search scheme to select optimal values of penalty parameters λ_1 and λ_2 for our DMCA method, with the results of the grid search shown in Fig. 4. Here, the value of each parameter is selected from the range of $[0.0001, 0.001, 0.01, 0.1, 1, 10]$. The OA index is exploited to measure the performance of our DMCA method on different parameter values. The MedHSI No. 015-01 data from the In Vivo HHB database are used as a testing example, and the analysis results can be generalized to the parameter section on other data.

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TABLE I
METRIC RESULTS OF DIFFERENT APPROACHES ON MEDHSI NO. 020-01 (MEDHSI-020-01) IN THE IN VIVO HHB DATASET

Class number/name	Methods								
	SVM	SVM-Gabor	MFASR	SAE	CNN	RPNet	FMDM	FUST	DMCA
1 Normal Tissue	87.62	97.38	96.79	94.12	**98.41**	98.14	97.86	96.77	**98.41**
2 Tumour Tissue	99.12	93.22	94.67	99.86	99.56	99.94	99.47	99.25	**99.97**
3 Hypervascularized Tissue	67.26	77.38	90.59	92.99	93.93	93.46	94.73	93.99	**99.93**
4 Labeled Background	89.51	94.30	96.65	99.27	**100**	**100**	99.54	99.92	99.81
OA(%)	89.30	91.83	94.97	97.53	98.57	98.60	98.44	98.13	**99.62**
AA(%)	85.88	90.57	94.67	96.56	97.97	97.88	97.90	97.48	**99.53**
Kappa×100	85.13	88.74	93.03	96.56	98.02	98.05	97.83	97.41	**99.48**

TABLE II
METRIC RESULTS OF DIFFERENT APPROACHES ON MEDHSI NO. 015-01 (MEDHSI-015-01) IN THE IN VIVO HHB DATASET

Class number/name	Methods								
	SVM	SVM-Gabor	MFASR	SAE	CNN	RPNet	FMDM	FUST	DMCA
1 Normal Tissue	83.78	95.81	98.05	99.52	97.34	93.62	97.39	95.35	**99.68**
2 Tumour Tissue	83.25	73.97	99.75	96.25	95.06	99.61	98.80	99.40	**99.80**
3 Hypervascularized Tissue	92.48	95.42	81.26	**99.95**	99.46	99.23	99.90	99.38	99.90
4 Labeled Background	60.89	71.43	98.41	95.51	95.22	97.83	**100**	99.27	97.10
OA(%)	86.07	88.00	89.95	98.56	97.65	98.34	99.24	98.75	**99.60**
AA(%)	80.10	84.16	94.37	97.81	96.77	97.57	99.02	98.35	**99.12**
Kappa×100	78.19	81.06	85.23	97.79	96.38	97.43	98.83	98.07	**99.38**

As can be found in Fig. 4, the performance (in OA) of the proposed DMCA method obviously fluctuates when changing the parameters λ_1 and λ_2, indicating that our proposed method is sensitive to parameter values. The best performance is achieved at $\lambda_1 = 0.1$ and $\lambda_2 = 0.01$.

D. Algorithm Comparison

In this section, we compare the proposed DMCA method with eight representative approaches, including SVM, SVM-Gabor [46], multiple-feature-based adaptive sparse representation (MFASR) [47], SAE [48], CNN [49], random patches network (RPNet) [50], FMDM [29], and FUST [30]. As can be known, the SVM method uses the original spectral pixels as discriminative features, and an SVM classifier is then adopted to predict labels. While for the SVM-Gabor, the Gabor features are first extracted to reflect the texture information, and then, an SVM is used for the label prediction. The MFASR is based on the sparse representation discrimination with multiple features (including original pixels, morphological profile, Gabor feature, and differential morphological profiles) for reporting the labels of MedHSIs. The SAE, CNN, RPNet, FMDM, and FUST are of advanced deep learning. For our DMCA method, the values of penalty parameters λ_1 and λ_2 are 0.1 and 0.01 according to the analysis results of parameter section in Section III-C, which are selected by the grid search scheme. The number of deep layers, i.e., $J+1$, is fixed to 4. In addition, the SVM is performed on official open source libsvm toolbox with optimal parameter setting. The parameter values of SVM-Gabor, MFASR, SAE, CNN, and RPNet are suggested by Hao et al. [29], Zeng et al. [30], Li and Du [46], Fang et al. [47], Chen et al. [48], Hu et al. [49], and Xu et al. [50].

1) Classification Comparison on Brain Cancer Recorded In Vivo HHB Dataset: We implemented our method and the eight competing approaches on the In Vivo HHB dataset, which records lots of human brain cancers. Due to the large amount of MediHSI data included in this dataset, two MediHSIs (i.e., MediHSI No. 020-01 and MediHSI No. 015-01) were randomly selected as representative examples for the algorithm comparison. Both MediHSIs have four labeled classes, including normal tissue, tumor tissue, hypervascularized tissue, and labeled background. In addition, to reduce computation burden and conveniently observe the difference among different methods, we used a 30 sampling factor to downsample MediHSI No. 020-01 and MediHSI No. 015-01, so that the number of spectral bands was reduced from 826 to 27. We selected 0.1% ratio of the number of pixels included in each class from each MediHSI as training samples, and the rest as testing samples. The experimental results and related discussion are presented in detail in the following.

We repeated the experiment ten times for each method and reported the averaged values of the OA, AA, and Kappa metrics as well as correct identification rate for each class in Tables I and II. As can be found in Tables I and II, the OA, AA, and Kappa results of the DMCA are consistently better than those of SVM, SVM-Gabor, MFASR, SAE, CNN, RPNet, FMDM, and FUST. Therefore, the DMCA achieves the higher classification accuracy on the whole, compared with other eight approaches. We can also see that our method reports the larger correct identification rates (i.e., 99.97% and 99.80%) on the tumor tissue class, which empirically validates its advantage in the diagnosis of brain cancers. In addition, the classification maps of different methods are

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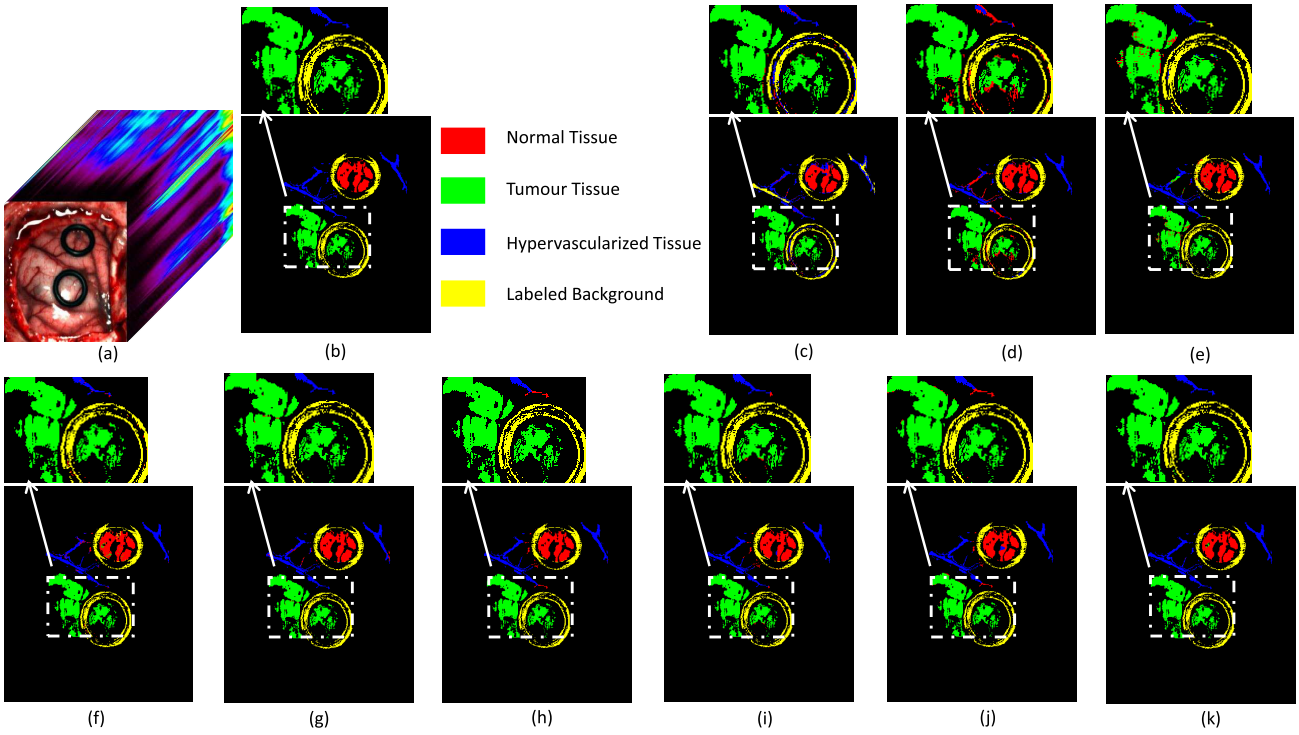


Fig. 5. Classification maps of different approaches on MedHSI No. 020-01 (MedHSI-020-01). (a) Input MedHSI. (b) Ground truth. (c) SVM. (d) SVM-Gabor. (e) MFASR. (f) SAE. (g) CNN. (h) RPNet. (i) FMDM. (j) FUST. (k) Proposed DMCA.

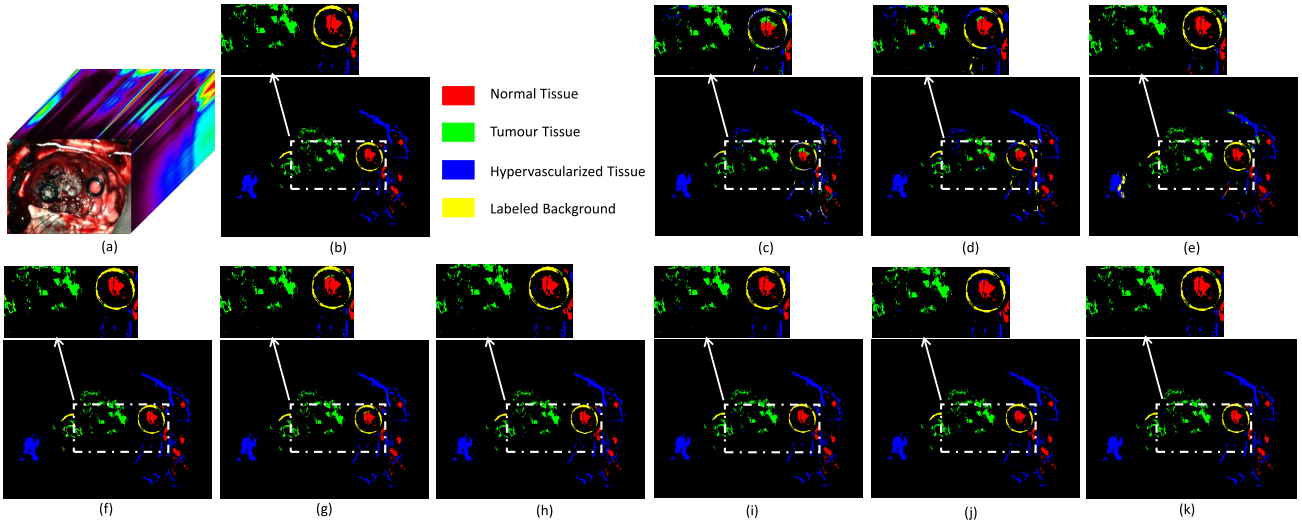


Fig. 6. Classification maps of different approaches on MedHSI No. 015-01 (MedHSI-015-01). (a) Input MedHSI. (b) Ground truth. (c) SVM. (d) SVM-Gabor. (e) MFASR. (f) SAE. (g) CNN. (h) RPNet. (i) FMDM. (j) FUST. (k) Proposed DMCA.

given in Figs. 5 and 6 to further validate the superiority of our method, where some parts are highlighted and enlarged by using white rectangles for clear observation. As can be found, the classification maps of our method in Figs. 5(k) and 6(k) are closer to the ground truths in Figs. 5(b) and 6(b) than those of SVM, SVM-Gabor, MFASR, SAE, CNN, RPNet, FMDM, and FUST in Figs. 5(c)–(j) and 6(c)–(j). According to the above descriptions, we conclude that our DMCA method obtains the more competitive performance than other eight competing approaches on the brain cancer recorded In Vivo HHB dataset.

2) Classification Comparison on Cholangiocarcinoma Recorded MC Dataset: This MC dataset records choledoch tissues from different patients, which is here used for the comparison between the proposed DMCA method with other eight advanced approaches. We randomly selected two MediHSI data as examples for the comparison evaluation, including both MediHSI No. x031368-20x-roi3 and MediHSI No. 033629-20x-roi5. The recorded image scenes have two kinds of regions, i.e., cholangiocarcinoma tissues and normal tissues. Here, because of the very high spatial resolution included in these two MediHSIs, in order to reduce

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TABLE III

METRIC RESULTS OF DIFFERENT APPROACHES ON MEDHSI NO. X031368-20X-ROI3 (MEDHSI-X031368-20X-ROI3) IN THE MC DATASET

Class number/name	Methods								
	SVM	SVM-Gabor	MFASR	SAE	CNN	RPNet	FMDM	FUST	DMCA
1 Cholangiocarcinoma Tissue	91.95	86.26	87.98	87.78	92.41	97.05	96.02	97.43	**99.88**
2 Labeled Norm Tissue	76.33	84.98	94.14	99.03	97.77	98.29	97.47	96.80	**99.62**
OA(%)	83.96	85.60	91.13	93.54	95.15	97.68	96.77	97.11	**99.75**
AA(%)	84.14	85.62	91.06	93.40	95.09	97.67	96.75	97.12	**99.75**
Kappa×100	68.02	71.21	82.23	87.03	90.29	95.36	93.52	94.22	**99.50**

TABLE IV

METRIC RESULTS OF DIFFERENT APPROACHES ON MEDHSI NO. 033629-3-20X-ROI5 (MEDHSI-033629-3-20X-ROI5) IN THE MC DATASET

Class number/name	Methods								
	SVM	SVM-Gabor	MFASR	SAE	CNN	RPNet	FMDM	FUST	DMCA
1 Cholangiocarcinoma Tissue	44.66	43.80	76.02	79.78	86.76	94.87	95.92	93.14	**99.48**
2 Labeled Norm Tissue	92.63	98.98	96.03	99.33	98.89	99.14	99.24	95.89	**99.84**
OA(%)	79.51	83.89	90.56	93.99	95.57	97.97	98.33	97.14	**99.74**
AA(%)	68.65	71.39	86.02	89.56	92.82	97.00	97.57	95.89	**99.66**
Kappa×100	41.97	51.35	75.19	83.95	88.48	94.84	95.76	92.72	**99.34**

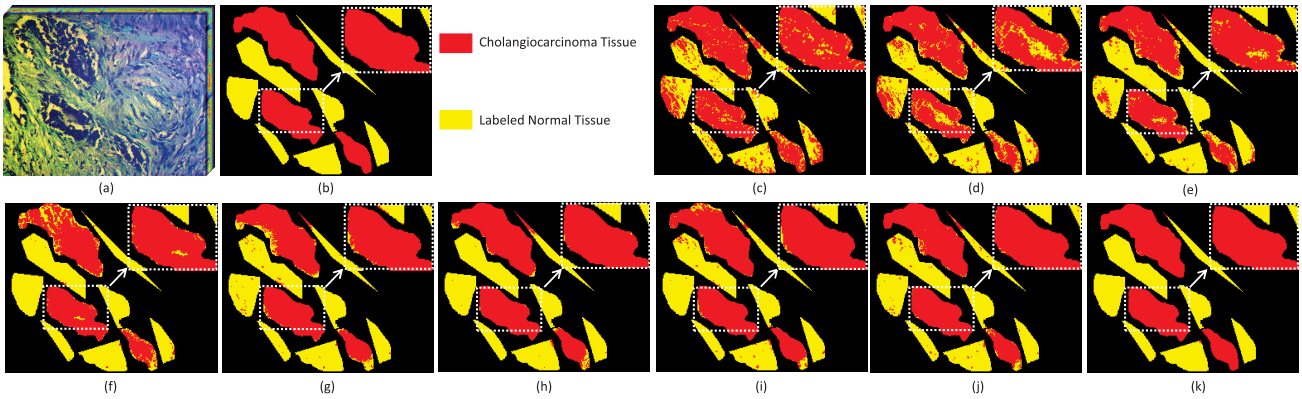


Fig. 7. Classification maps of different approaches on MedHSI No. x031368-20x-roi3 (MedHSI-x031368-20x-roi3). (a) Input MedHSI. (b) Ground truth. (c) SVM. (d) SVM-Gabor. (e) MFASR. (f) SAE. (g) CNN. (h) RPNet. (i) FMDM. (j) FUST. (k) Proposed DMCA.

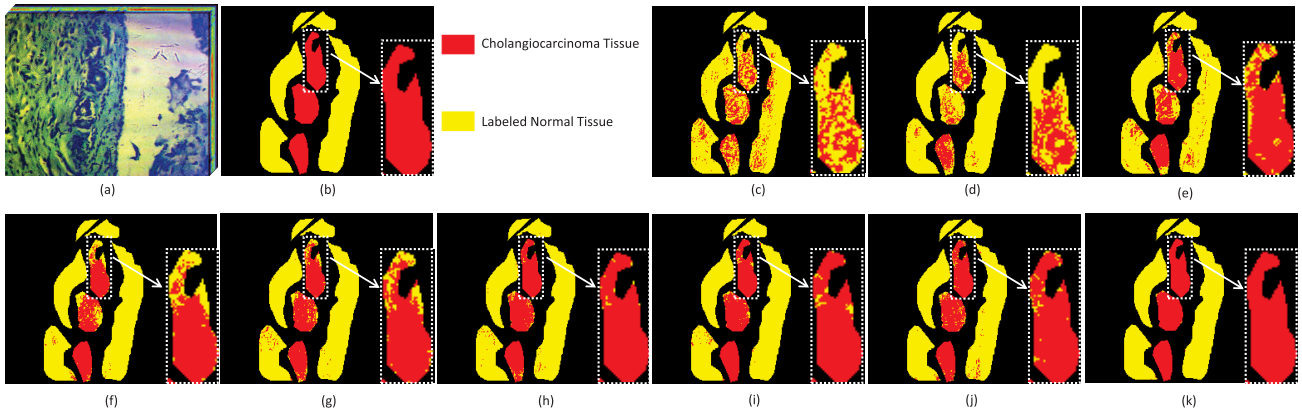


Fig. 8. Classification maps of different approaches on MedHSI No. 033629-3-20x-roi5 (MedHSI-033629-3-20x-roi5). (a) Input MedHSI. (b) Ground truth. (c) SVM. (d) SVM-Gabor. (e) MFASR. (f) SAE. (g) CNN. (h) RPNet. (i) FMDM. (j) FUST. (k) Proposed DMCA.

computational burden, partial regions of norm tissues were labeled, and a five sampling factor was used to downsample the spatial resolution of both MediHSIs, so that their spatial size was decreased from 1024 × 1280 to 205 × 256. We here chosen 0.2% of the amount of pixels in each class from each MedHSI as training samples, and the rest as testing samples.

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TABLE V
DMCA IS COMPARED WITH DMCA_NL AND DMCA_NC ON THE MEDHSI-020-01 AND MEDHSI-015-01 DATA, WHERE DMCA_NL DENOTES THE DMCA FORM WITHOUT THE MARGIN COSINE SOFT-MAX LOSS AND DMCA_NC DENOTES THE FORM WITHOUT COSINE MARGIN

Testing Data	Metrics	Methods		
		DMCA	DMCA_NL	DMCA_NC
MedHSI-020-01	OA(%)	99.62	96.53	99.19
	AA(%)	99.53	96.12	98.95
	Kappa×100	99.48	94.86	98.88
MedHSI-015-01	OA(%)	99.60	97.63	98.84
	AA(%)	99.12	95.88	97.79
	Kappa×100	99.38	96.34	98.21

The detailed experimental results and analysis can be seen as follows.

Similarly, we also repeated the experiment ten times for each method and listed the averaged OA, AA, and Kappa values as well as correct identification rate for each class in Tables III and IV. From these two tables, it can be directly found that the DMCA method achieves the high classification accuracy because of its larger OA, AA, and Kappa values. Also, the identification accuracy of cholangiocarcinoma obtained by our DMCA is higher than that of SVM, SVM-Gabor, MFASR, SAE, CNN, RPNet, FMDM, and FUST, which indicates that the stronger ability of cholangiocarcinoma diagnose is included in DMCA. Furthermore, as shown in Figs. 7 and 8, the classification maps of different approaches are provided and compared, and some parts are also highlighted by white rectangles for convenient comparison. We find that the proposed DMCA method in Figs. 7(k) and 8(k) achieves the closer maps to the ground truths in Figs. 7(b) and 8(b) compared with SVM, SVM-Gabor, MFASR, SAE, CNN, RPNet, FMDM, and FUST in Figs. 7(c)–(j) and 8(c)–(j). Therefore, it is concluded that the DMCA method outperforms other eight compared approaches on the cholangiocarcinoma recorded MC dataset.

E. Ablation Study

In order to comprehensively assess the performance of the proposed DMCA method, we here carry out the ablation study to explore the effectiveness of the embedded margin cosine soft-max loss and the effectiveness of the separate cosine margin m. Both MedHSI-020-01 and MedHSI-015-01 data from the In Vivo HHB dataset are here used for data support, and the OA, AA, and Kappa indexes are exploited to quantitatively measure results. The detailed study and discussion are shown in the following.

1) Effectiveness of Margin Cosine Soft-Max Loss: The margin cosine soft-max loss is introduced in the proposed DMCA method to have two roles, i.e., embracing the features that are more compatible with the classifier and maximizing the decision margin in the angular space. By such way, the extracted features can be easily and conveniently categorized. In order to validate the effectiveness of the margin cosine soft-max loss, we first remove this loss from the DMCA method. Thus, the form without margin cosine soft-max loss can be obtained, which is simply denoted by DMCA_NL. Then, the DMCA is compared with the DMCA_NL on MedHSI-020-01 and MedHSI-015-01 data, with the results visually shown in Table V. As can be found in this table, the OA, AA, and Kappa values of the DMCA are consistently bigger than those of the DMCA_NL on these two data, which indicates the effectiveness of the margin cosine soft-max loss.

2) Effectiveness of Cosine Margin: The cosine margin m is used in our DMCA to enlarge the discrimination of different feature clusters. For the sake of validating the effectiveness of such cosine margin, we first remove the margin from the DMCA method and acquire the form without cosine margin (denoted by DMCA_NC). Then, the DMCA is compared with the DMCA_NC on MedHSI-020-01 and MedHSI-015-01 data, and the results are presented in Table V. From this table, it can be observed that, the DMCA achieves the better OA, AA, and Kappa results compared with the DMCA_NC form. This implies that the used cosine margin m is effective in our DMCA method.

IV. CONCLUSION

In this study, we present a novel DMCA network to classify the MedHSIs for the sake of the tumor diagnosis. Different from traditional deep-learning-based approaches, our proposed DMCA method integrates the classifier for prediction into the extraction of deep features of MedHSIs, which can make the extracted features compatible with the classifier. Furthermore, a cosine margin is introduced into the built network, which aims to enhance the discrimination of different feature clusters. We evaluate the proposed DMCA method on two MedHSI datasets, i.e., In Vivo HHB database and MC database, with the results demonstrating its effectiveness and superiority in the classification and tumor diagnosis.

In practice, there exist many kinds of MedHSI datasets with different structural characteristics, such as brain cancer recorded In Vivo HHB dataset and cholangiocarcinoma recorded MC dataset. They were usually generated by different types of medical hyperspectral imaging equipment for different body parts. Therefore, it is infeasible to perform a cross-corpus study for our DMCA method. In the future, we will attempt to develop the improved DMCA form, which can be trained on one dataset and further applied to other datasets.

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